Fiscal Year:	FY 2022	**Task Last Updated:**	FY 03/12/2025
PI Name:	George, Stuart Ph.D.		
Project Title:	REM and CAD Radiation Monitoring for Private Astronaut Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	stuart.george@nasa.gov	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	281-244-1831
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2022	**End Date:**	03/31/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 03/31/2025 per TRISH (Ed., 3/12/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Garza, Ramona Ph.D. (Leidos Inc.) Stoffle, Nicholas Ph.D. (Leidos Inc.)		
Grant/Contract No.:	NNX16AO69A-EXP0011		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	"Enhancing eXploration Platforms and ANalog Definition" (EXPAND) Project This project outlines a set of radiation measurements aboard the Polaris Dawn and Axiom-2 commercial spaceflights. The ultimate goal of these measurements is to “better understand how space radiation affects human biological systems," but less prosaically is to measure: • Total mission absorbed doses and time resolved dose for crewmembers. • Time resolved dose, linear energy transfer, and proton spectra inside the capsule, with particular attention to events that can change radiation dose levels such as opening the capsule door and radiation belt transits.		

	This dataset is both unique and interesting in its own right. The planned 500 km orbit of the Polaris missions is higher than any since Apollo and includes Van Allen radiation belt transits, both factors make the radiation field quite different to the ISS environment that we are used to. This data is also highly relevant for other radiosensitive experiments that are flying on the Polaris Missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2022.
Bibliography Type:	Description: (Last Updated: 12/27/2024)